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旋转抛物面聚光效果的研究

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摘 要: 根据旋转抛物面的性质和光的反射定律, 本文讨论了位于焦点的点光源在测试屏上的照度分布、影响因素及点光源的位置对抛物面聚光作用的影响。

关键词: 点光源; 照度分布

分类号: AMS (2000) 49K35

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文献标识码: A

对理想点光源 (光学上称为朗伯特辐射体) 而言, 其光强 I 在以该光源为心的球面上均匀分布, 即在球面上各点沿法线方向其光强相等, 故通过球面上某给定区域的光通量 Φ 与该区域面积成正比。因此单位球面上的环带微元面积 ΔA 与纬度微元 $\Delta\theta$ 关系

$$\Delta A = 2\pi \cdot (\cos\theta - \cos(\theta + \Delta\theta))$$

$$\therefore \Delta\Phi = 2\pi\kappa(\cos\theta - \cos(\theta + \Delta\theta))$$

注 κ 是与点光源的光强成正比的正常数。

以抛物面焦点为球心, 作一含于抛物面的球, 由光的反射定律和旋转抛物面的对称性知, 由焦点出发通过图示环带区域的光线经抛物面反射后, 成为平行光照射到测试屏上形成一以光轴与测试屏交点 O' 为圆心的光环, 且同一圆周上各点照度相等, 由抛物面定义知抛物面上各点到焦点距离与到抛物面准面距离相等, 故由焦点出发到达测试屏的每束光线走过的总路程相等 (都等于准面到测试屏的距离) 且它们都与测试屏垂直。故通过测试屏上圆环的光通量与通过球面上环带的光通量相等。

由照度定义知: $E = d\Phi/dS$, 而 $\forall$ 测试屏上一点 y_θ , 包含该点的微圆环面积为 $dS = \pi \cdot (\|y_{\theta+d\theta}\|^2 - \|y_\theta\|^2)$, $\|y\|$ 定义为点 y 到光轴的距离。通过该微元的光通量 $d\Phi = 2\pi\kappa(\cos\theta - \cos(\theta + d\theta))$, 两者均只与 θ 角度及 $d\theta$ 有关, 故可将问题化到二维研究。为此, 将抛物面在二维时的抛物线方程写为以焦点 O 为极点的极坐标形式 (θ 角取由负极轴方向沿顺时针方向旋转所成的夹角为正) 如下

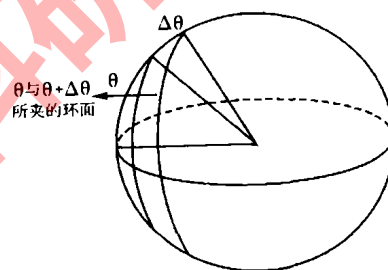


图 1

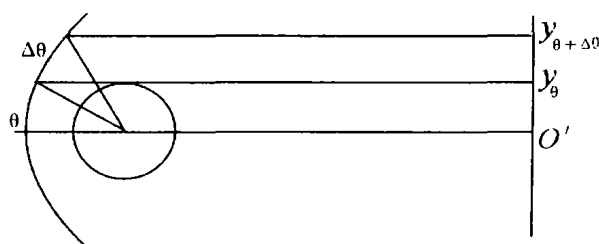


图 2

$$\begin{aligned}\rho &= p/(1 - \cos(\pi - \theta)) = p/(1 + \cos\theta) \\ y_\theta &= \rho \cdot \sin\theta = p \cdot \sin\theta/(1 + \cos\theta) \\ \Rightarrow y_\theta^2 &= p^2 \cdot \frac{\sin^2\theta}{(1 + \cos\theta)^2} = p^2 \cdot \frac{1 - \cos\theta}{1 + \cos\theta} = p^2 \cdot \left(\frac{2}{1 + \cos\theta} - 1\right) \\ \text{i.e. } 1 + \cos\theta &= \frac{2p^2}{y_\theta^2 + p^2} \\ y_{\theta+\Delta\theta}^2 &= p^2 \cdot \left(\frac{2}{1 + \cos(\theta + \Delta\theta)} - 1\right)\end{aligned}$$

$\therefore$ 圆环 $\odot_O(y_\theta, y_{\theta+\Delta\theta})$ 的面积为

$$\begin{aligned}\Delta S &= \pi \cdot (y_{\theta+\Delta\theta}^2 - y_\theta^2) = 2\pi p^2 \cdot \left(\frac{1}{1 + \cos(\theta + \Delta\theta)} - \frac{1}{1 + \cos\theta}\right) \\ &= 2\pi p^2 \cdot \frac{\cos\theta - \cos(\theta + \Delta\theta)}{(1 + \cos\theta) \cdot (1 + \cos(\theta + \Delta\theta))} \\ \therefore \frac{\Delta\Phi}{\Delta S} &= \frac{\kappa}{p^2} \cdot (1 + \cos\theta) \cdot (1 + \cos(\theta + \Delta\theta)) \\ \therefore \lim_{\Delta S \rightarrow 0} \frac{\Delta\Phi}{\Delta S} &= \frac{\kappa}{p^2} \cdot (1 + \cos\theta)^2 = \frac{\kappa}{p^2} \cdot \left(\frac{2p^2}{y_\theta^2 + p^2}\right)^2 = \frac{4\kappa p^2}{(y_\theta^2 + p^2)^2}\end{aligned}$$

若令 $y = tp, t \in R$ (显然可将 t 视为距光轴远近程度的度量), 则得到

$$E_r = \frac{4\kappa}{(t^2 + 1)^2} \cdot \frac{1}{p^2}$$

由此可见, 对理想点光源而言, 当其位于抛物面的焦点时, 在抛物面出口口径内, 测试屏上任意一点的照度 (如果不考虑直射光) 仅与它距光轴的远近程度和抛物面焦距长短有关。某点的照度与抛物面焦距的平方成反比关系, 且该点距光轴越远, 照度越低。实际上, 对理想点光源而言, 由直射光产生的照度与光源和被照点的距离的平方成反比。由于本题中测试距离远大于抛物面出口口径和焦距, 故直射光的影响可忽略不计。

以下讨论不在焦点处的理想点光源其纵坐标变化对聚光的影响。本文中对聚光的定义如下: 设点光源射出的光束夹角为 γ , 经抛物面反射后的夹角为 φ , 若 φ 始终小于 γ , 则认为抛物面对该点光源发出的光线有汇聚作用。

如图3所示, O 为抛物面焦点, 点光源 A 在焦平面上, AB, AC 为由点光源 A 发出的两条光线, 且分别交抛物面于 B, C 点, θ, β 为 B, C 点的极角的补角。本文仅讨论 $\theta \in (-\pi/2, \pi/2)$, 光源在焦点上方的情况。(因由对称性可知, 光源在焦点下方必得到相同的结论。)

设: $O(p, 0), A(p, \omega)$, 令 $\omega = t \cdot p, t \in (0, 1)$

$$|OB| = \rho_B, |OC| = \rho_C$$

$$\angle BOD = \theta, \angle COD = \beta, \angle BOC = \Delta\theta,$$

则: $\angle BAC = \gamma$, 由余弦定理易得

$$|AB|^2 = \rho_B^2 + \omega^2 - 2\rho_B \cdot \omega \cdot \sin\theta$$

$$|AC|^2 = \rho_C^2 + \omega^2 - 2\rho_C \cdot \omega \cdot \sin\beta$$

$$|BC|^2 = \rho_B^2 + \rho_C^2 - 2\rho_B \cdot \rho_C \cdot \cos\Delta\theta$$

由光路的可逆性知, 由 A 射出的光线 AB, AC 的反射线与水平线的夹角分别为 $\angle OBA, \angle OCA$, 故 $\angle OBA - \angle OCA = \varphi$

由三角函数性质有: $\cos\varphi = \cos\angle OBA \cdot \cos\angle OCA + \sin\angle OBA \cdot \sin\angle OCA$

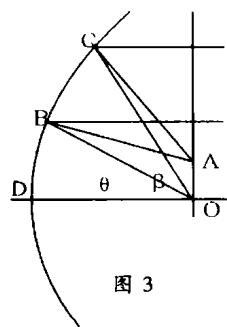


图3

又由余弦定理得

$$\begin{aligned}\cos\angle OBA &= \frac{|OB|^2 + |AB|^2 - |OA|^2}{2 \cdot |OB| \cdot |AB|} = \frac{2\rho_B^2 + \omega^2 - 2\rho_B\omega\sin\theta - \omega^2}{2\rho_B \cdot |AB|} \\ &= \frac{\rho_B - \omega \cdot \sin\theta}{|AB|}\end{aligned}$$

$$\cos\angle OCA = \frac{|OC|^2 + |AC|^2 - |OA|^2}{2 \cdot |OC| \cdot |AC|} = \frac{\rho_C - \omega \cdot \sin\beta}{|AC|}$$

$$\begin{aligned}\cos\angle BAC &= \cos\gamma = \frac{|AC|^2 + |AB|^2 - |BC|^2}{2 \cdot |AB| \cdot |AC|} \\ &= \frac{\omega^2 - \rho_B\omega \cdot \sin\theta - \rho_C\omega \cdot \sin\beta + \rho_B\rho_C\cos\Delta\theta}{|AB| \cdot |AC|}\end{aligned}$$

$$\because \theta, \beta \in (0, \pi/2) \quad \therefore \sin\angle OBA = \frac{\omega\cos\theta}{|AB|}, \sin\angle OCA = \frac{\omega\cos\beta}{|AC|}$$

$$\begin{aligned}\Rightarrow \cos\varphi &= \frac{(\rho_B - \omega\sin\theta)(\rho_C - \omega\sin\beta) + \omega^2\cos\theta\cos\beta}{|AB| \cdot |AC|} \\ &= \frac{\rho_B\rho_C - \rho_B\omega\sin\beta - \rho_C\omega\sin\theta + \omega^2\cos\Delta\theta}{|AB| \cdot |AC|}\end{aligned}$$

设 $|AB| \cdot |AC| = 1/k > 0$, 则有

$$\cos\varphi - \cos\gamma = k \cdot [(\rho_B\rho_C - \omega^2)(1 - \cos\Delta\theta) - (\sin\beta - \sin\theta)(\rho_B - \rho_C)\omega]$$

$$\therefore \rho = \frac{p}{1 + \cos\theta}, \omega = t \cdot p$$

$$\begin{aligned}\Rightarrow \frac{\cos\varphi - \cos\gamma}{kp^2} &= \frac{1}{p^2} \left[\left(\frac{p}{1 + \cos\theta} \frac{p}{1 + \cos\beta} - t^2 p^2 \right) (1 - \cos\Delta\theta) - (\sin\beta - \sin\theta) \left(\frac{p}{1 + \cos\theta} - \frac{p}{1 + \cos\beta} \right) t p \right] \\ &= \left(\frac{1}{(1 + \cos\theta)(1 + \cos\beta)} - t^2 \right) (1 - \cos\Delta\theta) - \frac{(\sin\beta - \sin\theta)(\cos\beta - \cos\theta)}{(1 + \cos\beta)(1 + \cos\theta)} \cdot t \\ &= - (1 - \cos\Delta\theta) t^2 - \frac{(\sin\beta - \sin\theta)(\cos\beta - \cos\theta)}{(1 + \cos\beta)(1 + \cos\theta)} t + \frac{1 - \cos\Delta\theta}{(1 + \cos\theta)(1 + \cos\beta)} \\ &\triangleq at^2 + bt + c = f(t)\end{aligned}$$

对此关于 t 的二次三项式, 显然 $a < 0, \Delta > 0, c > 0, b > 0$ 曲线开口向下。设 t^+, t^- 分别为 $f(t) = 0$ 的正、负根。易知 $t \in (0, t^+)$ 时, $f(t) > 0$, 即 $\varphi < \gamma$, 此时反射光夹角比出射光夹角小, 抛物面对点光源起聚光作用。

$$\begin{aligned}t^+ &= \frac{\sqrt{\left(\frac{(\sin\beta - \sin\theta)(\cos\beta - \cos\theta)}{(1 + \cos\beta)(1 + \cos\theta)} \right)^2 + \frac{4(1 - \cos\Delta\theta)^2}{(1 + \cos\theta)(1 + \cos\beta)}} - \frac{(\sin\beta - \sin\theta)(\cos\beta - \cos\theta)}{(1 + \cos\beta)(1 + \cos\theta)}}{2(1 - \cos\Delta\theta)} \\ &= \frac{\sqrt{M^2 + 4N \cdot L^2} - M}{2L \cdot N} \quad L \triangleq 1 - \cos\Delta\theta\end{aligned}$$

$$M \triangleq (\sin\beta - \sin\theta)(\cos\beta - \cos\theta), \quad N \triangleq (1 + \cos\beta)(1 + \cos\theta)$$

用 Matlab 绘出一系列 $\Delta\theta$ 下的 $t^+ - \theta$ 曲线, 得到 $t^+ \geq 0.42$, 而当 $\Delta\theta$ 足够小 (如 $\Delta\theta \leq 1.0537 \times 10^{-3}$) 时, 则有 $t^+ \geq 0.47$ 。

与用 Matlab 编程计算求得的如下 $\text{Min}(t^+)$ 基本一致。

$$\Delta\theta = 0.1, 0.01, 0.001, 0.0000005, 0.000000011357$$

$$\text{Min}(t^*) = 0.4255, 0.4253, 0.4253, 0.4253, 0.4669$$

于是我们可以断定,当点光源纵坐标位于 $-0.425p$ 到 $0.425p$ 之间时,在过光轴和点光源的平面内总是聚光的。而当点光源纵坐标位于 $-0.425p$ 和 $0.425p$ 之外时,在过光轴和点光源的平面内是否聚光与该光束的仰角有关。

当点光源不在焦点位置时,以上实际上只讨论了过光轴和点光源的平面内光的反射情况,并得到简洁的结果。这是因为在此时,各入射、反射光线均共面,易于处理。但若考虑向任意方向的光线的反射情况,则情况复杂得多,首先反射光线几乎都不共面,其次参数变量增加,表达式异常复杂,仍沿用上方方法难于处理。在此情况下,我们首先建立以光源点为心的单位球面,使用 Mathematics 计算了该球面上的点映射到测试平面上时的 Jacobi 值 J 。再利用在球面上光强的均匀分布性质,则可得到测试屏上光强度。

设点光源 P 坐标 (x_p, y_p, z_p) , 抛物面上反射点 $Q(x_q, y_q, z_q)$, 反射光与测试屏的交点 $R(x_r, y_r, z_r)$ 。(其中 $x_p = 0, z_p = 15, z_r = 25015$)

则 Q 点处法向量为 $\vec{n} = (x_q/30, y_q/30, -1)$; PQ 对应向量为 $\vec{a} = ((x_q - x_p), (y_q - y_p), (x_q^2 + y_q^2)/60 - z_p)$; QR 对应向量为 $\vec{b} = ((x_r - x_q), (y_r - y_q), -(x_q^2 + y_q^2)/60 + z_r)$;

由: $\vec{b} = k \cdot (\vec{a} - 2\vec{n} \cdot \frac{\vec{a} \cdot \vec{n}}{\|\vec{n}\|^2})$ 得到

$$x_r = -(x_p \cdot (x_q^4 - (900 + y_q^2)(y_q^2 - 60z_r) - 60x_q^2(-45 + z_r)) - x_q \cdot (x_q^4 + 1800y_q^2 + y_q^4 + 2x_q^2(900 - y_p y_q + y_q^2) + 54000z_p - 2y_p y_q(1800 + y_q^2 - 60z_r) + 54000z_r - 3600z_p z_r) / (-3600x_p x_q + x_q^4 - 3600y_p y_q + 2700y_q^2 + y_q^4 + 2x_q^2(1350 + y_q^2 - 30z_p) + 54000z_p - 60y_q^2 z_p)$$

$$y_r = -(2x_p x_q^3 y_q - x_q^4 (y_p + y_q) - y_q (1800y_q^2 + y_q^4 - 3600(z_p(z_r - 15) - 15z_r)) + 2x_p x_q y_q (y_q^2 - 60(z_r - 30)) - 2x_q^2 (y_q (900 + y_q^2) - 30y_p (z_r - 15)) + y_p (y_q^4 - 60y_q^2 (z_r - 45) + 54000z_r)) /$$

$$(-3600x_p x_q + x_q^4 - 3600y_p y_q + 2700y_q^2 + y_q^4 + 2x_q^2(1350 + y_q^2 - 30z_p) + 54000z_p - 60y_q^2 z_p)$$

得到光源在 (x_p, y_p, z_p) 时, $(x_r, y_r) \rightarrow (x_q, y_q)$ 时偏导数矩阵(分号表示行的结束,逗号表示列的间隔)如下

$$\begin{aligned} & (x_q^8 + 7200y_p^2 y_q^2 (-1499100 + y_q^2) - 2y_p y_q (-1497300 + y_q^2)(900 + y_q^2)^2 + (900 + y_q^2)^4 \\ & + 2x_q^6 (y_p y_q + 2(900 + y_q^2)) + 2x_q^4 (y_p y_q (-4508100 + y_q^2) + 3(900 + y_q^2)^2) \\ & + 2x_q^2 (10800y_p^2 y_q^2 + 2(900 + y_q^2)^3 - y_p y_q (2710530000 + 3012600y_q^2 + y_q^4))) / \\ & (x_q^4 - 3600y_p y_q + 2x_q^2(900 + y_q^2) + (900 + y_q^2)^2)^2, \\ & (2x_q y_p (1215729000000 - x_q^6 - 2704050000y_q^2 + 7200y_p y_q^3 - 4504500y_q^4 + y_q^6 \\ & - x_q^4 (-1499100 + y_q^2) + x_q^2 (2700810000 - 3005400y_q^2 + y_q^4))) / \\ & (x_q^4 - 3600y_p y_q + 2x_q^2(900 + y_q^2) + (900 + y_q^2)^2)^2; \\ & 4x_q y_p (608229000000 + 2700000000y_p y_q - 1350810000y_q^2 - 2250900y_q^4 + y_q^6 \\ & + x_q^4 (750900 + y_q^2) - 2x_q^2 (-675810000 + 1800y_p y_q + 750000y_q^2 - y_q^4))) / \\ & (x_q^4 - 3600y_p y_q + 2x_q^2(900 + y_q^2) + (900 + y_q^2)^2)^2, \\ & (x_q^8 + 4x_q^6 (900 - y_p y_q + y_q^2) + 2x_q^4 (1800y_p^2 - 4y_p y_q (-1124100 + y_q^2) + 3(900 + y_q^2)^2) \\ & + (900 + y_q^2)(10800y_p^2 (y_q^2 - 500300) + (900 + y_q^2)^3 - 19200y_p y_q (157y_q^2 - 421875)) \\ & + 4x_q^2 (y_p (4049190000y_q + 1494600y_q^3 - y_q^5) + (900 + y_q^2)^3 - 1350000000y_p^2)) / \\ & (x_q^4 - 3600y_p y_q + 2x_q^2(900 + y_q^2) + (900 + y_q^2)^2)^2 \end{aligned}$$



此行列式值 J_1 为

$$\begin{aligned} & ((900 + x_q^2 + y_q^2)(x_q^{10} + (900 + y_q^2)^5 - 21600y_p^3y_q(749999730000 - 1999400y_q^2 + y_q^4) \\ & - 2y_p y_q(900 + y_q^2)^2(y_q^4 + 9000y_q^2 - 5399190000) + x_q^8(5(900 + y_q^2) - 2y_p y_q) \\ & + 1200y_p^2(5033y_q^6 - 7549464900y_q^4 + 20229752430000y_q^2 - 3647187000000) + \\ & 2x_q^6(3005400y_p^2 + 5(900 + y_q^2)^2 - 4y_p y_q(2700 + y_q^2)) - 2x_q^4(10800y_p^3y_q \\ & - 5(900 + y_q^2)^3 - 1800y_p^2(5017y_q^2 - 2504499100) + 6y_p y_q(y_q^4 + 5400y_q^2 - 897030000)) \\ & + x_q^2(-43200y_p^3y_q(y_q^2 - 1000300) + 90000y_p^2(201y_q^4 - 200839496y_q^2 - 90270097200) \\ & + 5(900 + y_q^2)^4 - 8y_p y_q(y_q^6 + 8100y_q^4 - 2691090000y_q^2 - 2427813000000))) / \\ & (x_q^4 - 3600y_p y_q + 2x_q^2(900 + y_q^2) + (900 + y_q^2)^2)^3) \end{aligned}$$

若将 Q 取以 P 为心的球坐标 (φ, θ, R) , 易得球坐标与直角坐标间关系式

$$\varphi = \arctan \frac{y_q - y_p}{x_q - x_p}, \theta = \arccos \frac{z_q - z_p}{\|PQ\|}$$

得到 $(\theta, \varphi) \rightarrow (x_q, y_q)$ 的偏导数矩阵

$$\begin{aligned} & \frac{y_p - y_q}{x_q^2 + (y_p - y_q)^2}, \frac{x_q}{x_q^2 + (y_p - y_q)^2}; \\ & - \frac{60x_q(900 + x_q^2 + 2y_p^2 - 4y_p y_q + y_q^2)/\sqrt{x_q^2 + (y_p - y_q)^2}}{x_q^4 + 3600y_p^2 - 7200y_p y_q + 2x_q^2(900 + y_q^2) + (900 + y_q^2)^2}, \\ & - \frac{60(2y_p^2 y_q + y_q(900 + x_q^2 + y_q^2) + y_p(x_q^2 - 900 - 3y_q^2))/\sqrt{x_q^2 + (y_p - y_q)^2}}{x_q^4 + 3600y_p^2 - 7200y_p y_q + 2x_q^2(900 + y_q^2) + (900 + y_q^2)^2} \end{aligned}$$

此行列式值 J_2 为

$$60 \cdot \frac{(900 + x_q^2 - 2y_p y_q + y_q^2)/\sqrt{x_q^2 + (y_p - y_q)^2}}{x_q^4 + 3600y_p^2 - 7200y_p y_q + 2x_q^2(900 + y_q^2) + (900 + y_q^2)^2}$$

则由 $(\theta, \varphi) \rightarrow (x_r, y_r)$ 的变换的 Jacobi 值为 $J = J_2/J_1$

$$\begin{aligned} & (60 \cdot (900 + x_q^2 - 2y_p y_q + y_q^2) \cdot (x_q^4 - 3600y_p y_q + 2x_q^2(900 + y_q^2) + (900 + y_q^2)^2)^3) / \\ & ((900 + x_q^2 + y_q^2) \cdot (x_q^4 + 3600y_p^2 - 7200y_p y_q + 2x_q^2(900 + y_q^2) + (900 + y_q^2)^2) \\ & \sqrt{x_q^2 + (y_p - y_q)^2} \cdot (x_q^{10} + (900 + y_q^2)^5 - 21600y_p^3y_q(749999730000 - 1999400y_q^2 + y_q^4) \\ & - 2y_p y_q \cdot (900 + y_q^2)^2 \cdot (y_q^4 + 9000y_q^2 - 5399190000) + x_q^8 \cdot (4500 - 2y_p y_q + 5y_q^2) \\ & + 1200y_p^2 \cdot (5033y_q^6 - 7549464900y_q^4 + 20229752430000y_q^2 - 3647187000000) \\ & + 2x_q^6 \cdot (3005400y_p^2 + 5(900 + y_q^2)^2 - 4y_p y_q(2700 + y_q^2)) - 2x_q^4 \cdot (10800y_p^3y_q - 5(900 + y_q^2)^3 \\ & - 1800y_p^2(5017y_q^2 - 2504499100) + 6y_p y_q(y_q^4 + 5400y_q^2 - 897030000)) \\ & - x_q^2 \cdot (43200y_p^3y_q(y_q^2 - 1000300) - 5(900 + y_q^2)^4 - 90000y_p^2(201y_q^4 - 200839496y_q^2 - 90270097200) \\ & + 8y_p y_q(y_q^6 + 8100y_q^4 - 2691090000y_q^2 - 2427813000000))) \end{aligned}$$

由于分母可以为 0, J 的值为 ∞ , 因而抛物面对不在焦点的点光源发出的部分光线也有强烈的聚焦作用。

The Study of Paraboloidal Mirror for Linear Headlight Source

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